Analysis Tutorial 7

Problem 19

(i) First let f be a non-negative, simple function.

Then f is δ_ω-integrable if $\int_\Omega f \, d\delta_\omega < \infty$.

We have $\int_\Omega f \, d\delta_\omega = \sum_{x \in f(\Omega)} x \, \delta_\omega(f = x)$

Now $\delta_\omega(f = x) = 1 \iff \omega \in \{\omega' \in \Omega : f(\omega') = x\}$

$\iff f(\omega) = x$

So $\sum_{x \in f(\Omega)} x \, \delta_\omega(f = x) = f(\omega)$

(ii) Now let f be a non-negative, measurable function

Then $\int_\Omega f \, d\delta_\omega = \sup\{\int g \, d\delta_\omega : g : \Omega \to \mathbb{R}_+ \text{ is simple}, g \leq f\}$

Since $f \geq 0$, we can assume $g \geq 0$. In this case, $\int g \, d\delta_\omega = g(\omega)$.

So $\int_\Omega f \, d\delta_\omega = \sup\{g(\omega) : g : \Omega \to \mathbb{R}_+ \text{ is simple}, g \leq f\} \leq f(\omega)$.

But define $g(x) = \begin{cases} f(x) & x = \omega \\ 0 & \text{otherwise} \end{cases}$ is simple, so $\int_\Omega f \, d\delta_\omega \geq f(\omega) \Rightarrow \int_\Omega f \, d\delta_\omega = f(\omega)$

(iii) Let f be an arbitrary real-valued measurable function.

Then $\int_\Omega f \, d\delta_\omega = \underbrace{\int_\Omega f^+ \, d\delta_\omega}_{<\infty} - \underbrace{\int f^- \, d\delta_\omega}_{<\infty}$ by (ii)

We know $\int_\Omega f^+ \, d\delta_\omega = f^+(\omega)$

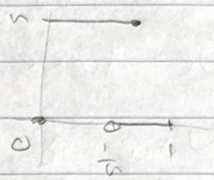
$\int_\Omega f^- \, d\delta_\omega = f^-(\omega)$

So $\int_\Omega f \, d\delta_\omega = f^+(\omega) - f^-(\omega) = f(\omega)$

Problem 20

Define $f_n : [0,1] \to \overline{\mathbb{R}}_+$ as

$$f_n(x) = \begin{cases} n & 0 \le x \le \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$



with the Borel σ-algebra and Lebesgue measure

Then $\int_{[0,1]} f_n \, d\lambda = 1$ for every $n \in \mathbb{N}$, and thus $\liminf_{n\to\infty} \int_{[0,1]} f_n \, d\lambda = 1$

But $\liminf_{n\to\infty} f_n(x) = 0$, so $\int_{[0,1]} \liminf_{n\to\infty} f_n \, d\lambda = 0$

Problem 21

Since $e^{-\alpha|x|} f(x) \le f(x)$ for $\alpha > 0$,

$$\int_{\mathbb{R}} e^{-\alpha|x|} f(x) \, dx \le \int_{\mathbb{R}} f(x) \, dx < \infty$$

So $\lim_{\alpha\to 0^+} \int_{\mathbb{R}} e^{-\alpha|x|} f(x) \, dx$ exists, and

so $$\lim_{\alpha\to 0^+} \int_{\mathbb{R}} e^{-\alpha|x|} f(x) \, dx = \lim_{n\to\infty} \int_{\mathbb{R}} e^{-\frac{|x|}{n}} f(x) \, dx$$

Let $f_n(x) = e^{-\frac{|x|}{n}} f(x)$. Then $\lim_{n\to\infty} f_n(x) = f(x)$, and $|f_n| \le |f|$ which is integrable, so

$$\lim_{n\to\infty} \int_{\mathbb{R}} e^{-\frac{|x|}{n}} f(x) \, dx = \int_{\mathbb{R}} f(x) \, dx \quad [\text{LBCT}]$$